

Template

Studentnames and studentnumbers here

2026-06-12

Set-up your environment

Title Page

Include your names Melvin Hofman Max Bergsma Gijs Stevens Hidde Voors Imad el Hilali

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Housing affordability has become an increasingly important social issue in the Netherlands over the past decade. House prices have risen substantially in many parts of the country, while income growth may not have kept pace. As a result, many households, particularly first-time buyers and younger generations, face greater difficulties entering the housing market. Rising housing costs can contribute to financial stress, wealth inequality, and reduced access to homeownership.

Understanding whether house prices have increased faster than incomes is therefore important for evaluating changes in housing affordability. If housing prices grow more rapidly than household incomes, purchasing a home becomes increasingly difficult for a large share of the population. This may have significant social and economic consequences, including increased inequality between homeowners and non-homeowners.

The main research question of this study is:

Have housing prices in the Netherlands increased faster than incomes over the past ten years?

To answer this question, the following sub-questions are examined: 1. How have house prices and average incomes developed between 2014 and 2024? 2. How has housing affordability changed during this period? 3. Are there significant differences in housing affordability between Dutch provinces? 4. Did the COVID-19 pandemic influence the relationship between house prices and income?

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
# data_download <- cbs_get_data("70072NED")

# raw_dataset <- data_download %>% cbs_add_label_columns()

# write.csv(raw_dataset, "data/raw_dataset.csv")
raw_dataset = read.csv("data/raw_dataset.csv")
```

2.2 Provide a short summary of the dataset(s)

```
huizenprijzen_provincie = raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2023,
    startsWith(RegioS, "PV")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    GemiddeldeWOZWaardeVanWoningen_98
  )
write.csv(huizenprijzen_provincie, "data/huizenprijzen_provincie.csv")

head(huizenprijzen_provincie)
```

##	Year	RegioS	RegioS_label	GemiddeldeWOZWaardeVanWoningen_98
## 1	2023	PV20	Groningen (PV)	268
## 2	2023	PV21	Fryslân (PV)	277
## 3	2023	PV22	Drenthe (PV)	294
## 4	2023	PV23	Overijssel (PV)	325
## 5	2023	PV24	Flevoland (PV)	347
## 6	2023	PV25	Gelderland (PV)	364

```
inkomen_provincie <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2023,
    startsWith(RegioS, "PV")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    BronInkomenAlsWerknemer_141
  )
write.csv(inkomen_provincie, "data/inkomen_provincie.csv")

head(inkomen_provincie)
```

##	Year	RegioS	RegioS_label	BronInkomenAlsWerknemer_141
----	------	--------	--------------	-----------------------------

```
## 1 2023 PV20    Groningen (PV)          37.5
## 2 2023 PV21    Fryslân (PV)           37.9
## 3 2023 PV22    Drenthe (PV)           39.7
## 4 2023 PV23    Overijssel (PV)        39.5
## 5 2023 PV24    Flevoland (PV)         40.4
## 6 2023 PV25    Gelderland (PV)        40.6
```

```
provincie_samengevoegd = huizenprijzen_provincie %>%
  inner_join(
    inkomen_provincie,
    by = c("Year", "RegioS", "RegioS_label")
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(statcode = RegioS)
head(provincie_samengevoegd)
```

```
##   Year RegioS   RegioS_label GemiddeldeWOZ InkomenWerknemer statcode
## 1 2023 PV20    Groningen (PV)          268          37.5    PV20
## 2 2023 PV21    Fryslân (PV)           277          37.9    PV21
## 3 2023 PV22    Drenthe (PV)           294          39.7    PV22
## 4 2023 PV23    Overijssel (PV)        325          39.5    PV23
## 5 2023 PV24    Flevoland (PV)        347          40.4    PV24
## 6 2023 PV25    Gelderland (PV)        364          40.6    PV25
```

```
provincie_summary = provincie_samengevoegd %>%
  summarise(
    Mean_WOZ = mean(GemiddeldeWOZ),
    Min_WOZ = min(GemiddeldeWOZ),
    Max_WOZ = max(GemiddeldeWOZ),
    SD_WOZ = sd(GemiddeldeWOZ),

    Mean_Income = mean(InkomenWerknemer),
    Min_Income = min(InkomenWerknemer),
    Max_Income = max(InkomenWerknemer),
    SD_Income = sd(InkomenWerknemer),
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = c("Statistic", "Variable"),
    names_sep = "_",
    values_to = "Value"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = Value
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ = "Average WOZ Value",
      Income = "Employee Income",
      Affordability = "WOZ-to-Income Ratio"
    )
  )
```

```

)
) %>%
select(Variable, Mean, Min, Max, SD)

kable(
  provincie_summary,
  caption = "Descriptive statistics for home value in provinces in the Netherlands, 2023",
  digits = 2
)

```

Table 1: Descriptive statistics for home value in provinces in the Netherlands, 2023

Variable	Mean	Min	Max	SD
Average WOZ Value	340.00	268.0	460.0	64.25
Employee Income	40.76	37.5	44.7	2.24

```

provincie_samenveoged <- provincie_samenveoged %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )
provincie_samenveoged

```

```

##   Year RegioS      RegioS_label GemiddeldeWOZ InkomenWerknemer statcode
## 1  2023 PV20      Groningen (PV)          268           37.5      PV20
## 2  2023 PV21      Fryslân (PV)          277           37.9      PV21
## 3  2023 PV22      Drenthe (PV)          294           39.7      PV22
## 4  2023 PV23      Overijssel (PV)        325           39.5      PV23
## 5  2023 PV24      Flevoland (PV)         347           40.4      PV24
## 6  2023 PV25      Gelderland (PV)        364           40.6      PV25
## 7  2023 PV26      Utrecht (PV)          447           44.4      PV26
## 8  2023 PV27      Noord-Holland (PV)      460           44.7      PV27
## 9  2023 PV28      Zuid-Holland (PV)      359           42.4      PV28
## 10 2023 PV29      Zeeland (PV)          281           40.3      PV29
## 11 2023 PV30      Noord-Brabant (PV)      370           41.8      PV30
## 12 2023 PV31      Limburg (PV)          288           39.9      PV31
##   AffordabilityRatio
## 1              7.146667
## 2              7.308707
## 3              7.405542
## 4              8.227848
## 5              8.589109
## 6              8.965517
## 7             10.067568
## 8             10.290828
## 9              8.466981
## 10             6.972705
## 11             8.851675
## 12             7.218045

```

2e Nieuwe variable

```

write.csv(provincie_samenveoged, "data/provincie_samenveoged.csv")

```

```

landelijk <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year >= 2015,
    Year <= 2024,
    startsWith(RegioS, "NL")
  ) %>%
  select(
    Year,
    RegioS_label,
    BronInkomenAlsWerknemer_141,
    GemiddeldeWOZWaardeVanWoningen_98
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer,
    WOZ_growth_yoy = (GemiddeldeWOZ / lag(GemiddeldeWOZ) - 1) * 100,
    Income_growth_yoy = (InkomenWerknemer / lag(InkomenWerknemer) - 1) * 100,
    GrowthGap_yoy = WOZ_growth_yoy - Income_growth_yoy
  )

write.csv(landelijk, "data/landelijk.csv")

```

landelijk

##	Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ	AffordabilityRatio
## 1	2015	Nederland	30.3	206	6.798680
## 2	2016	Nederland	31.4	208	6.624204
## 3	2017	Nederland	32.0	217	6.781250
## 4	2018	Nederland	32.3	230	7.120743
## 5	2019	Nederland	33.7	250	7.418398
## 6	2020	Nederland	35.3	271	7.677054
## 7	2021	Nederland	37.1	290	7.816712
## 8	2022	Nederland	39.0	317	8.128205
## 9	2023	Nederland	41.8	368	8.803828
## 10	2024	Nederland	44.7	378	8.456376
##	WOZ_growth_yoy	Income_growth_yoy	GrowthGap_yoy		
## 1	NA	NA	NA		
## 2	0.9708738	3.630363	-2.659489		
## 3	4.3269231	1.910828	2.416095		
## 4	5.9907834	0.937500	5.053283		
## 5	8.6956522	4.334365	4.361287		
## 6	8.4000000	4.747774	3.652226		
## 7	7.0110701	5.099150	1.911920		
## 8	9.3103448	5.121294	4.189051		
## 9	16.0883281	7.179487	8.908841		
## 10	2.7173913	6.937799	-4.220408		

```

provincie_map <- cbs_get_sf("provincie", 2023, verbose = TRUE)

## Note that `cbs_add_statcode` can be used for connecting the map to the data.
## Retrieving list of geojson files: 'https://cartomap.github.io/nl/index.csv'
## Reading layer `provincie_2023' from data source
## `https://cartomap.github.io/nl/rd/provincie_2023.geojson' using driver `GeoJSON'
## Simple feature collection with 12 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 13565 ymin: 306960 xmax: 277529 ymax: 619172
## Projected CRS: Amersfoort / RD New

provincie_samenveogd <- provincie_samenveogd %>%
  mutate(
    statcode = trimws(RegioS),
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
  )

provincie_map <- provincie_map %>%
  mutate(
    statcode = trimws(statcode)
  )

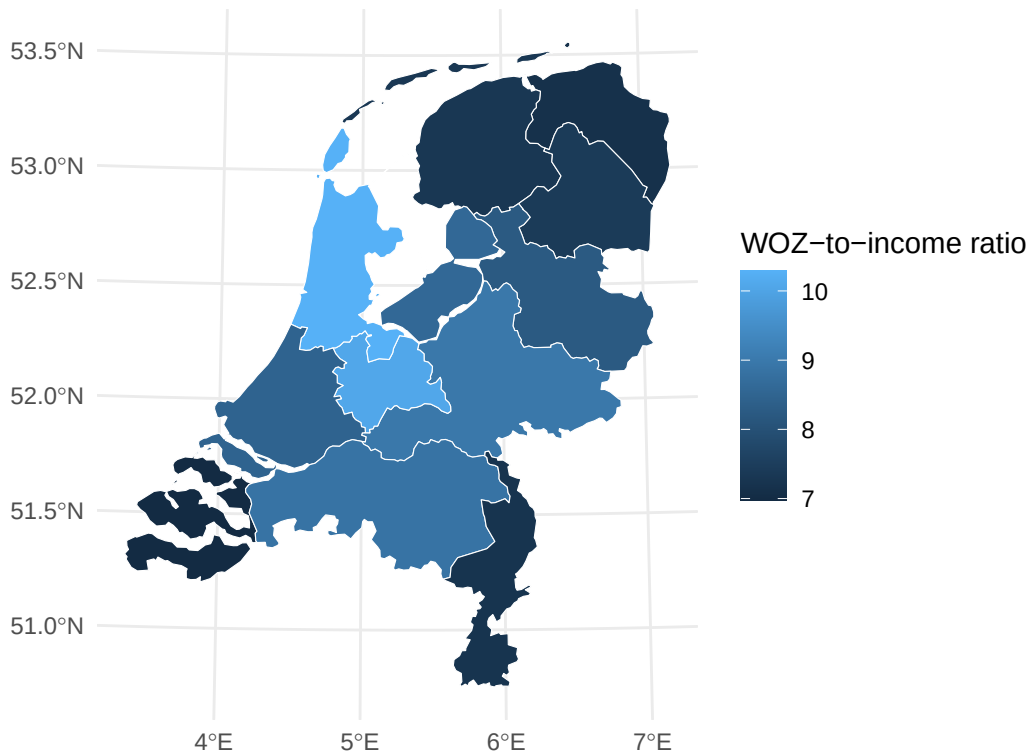
provincie_map_data <- provincie_map %>%
  left_join(
    provincie_samenveogd,
    by = "statcode"
  )

ggplot(provincie_map_data) +
  geom_sf(aes(fill = AffordabilityRatio), color = "white") +
  labs(
    title = "Housing Affordability by Province in 2023",
    subtitle = "Average WOZ value divided by employee income",
    fill = "WOZ-to-income ratio"
  ) +
  theme_minimal()

```

Housing Affordability by Province in 2023

Average WOZ value divided by employee income



Deze plot gemaakt met chat

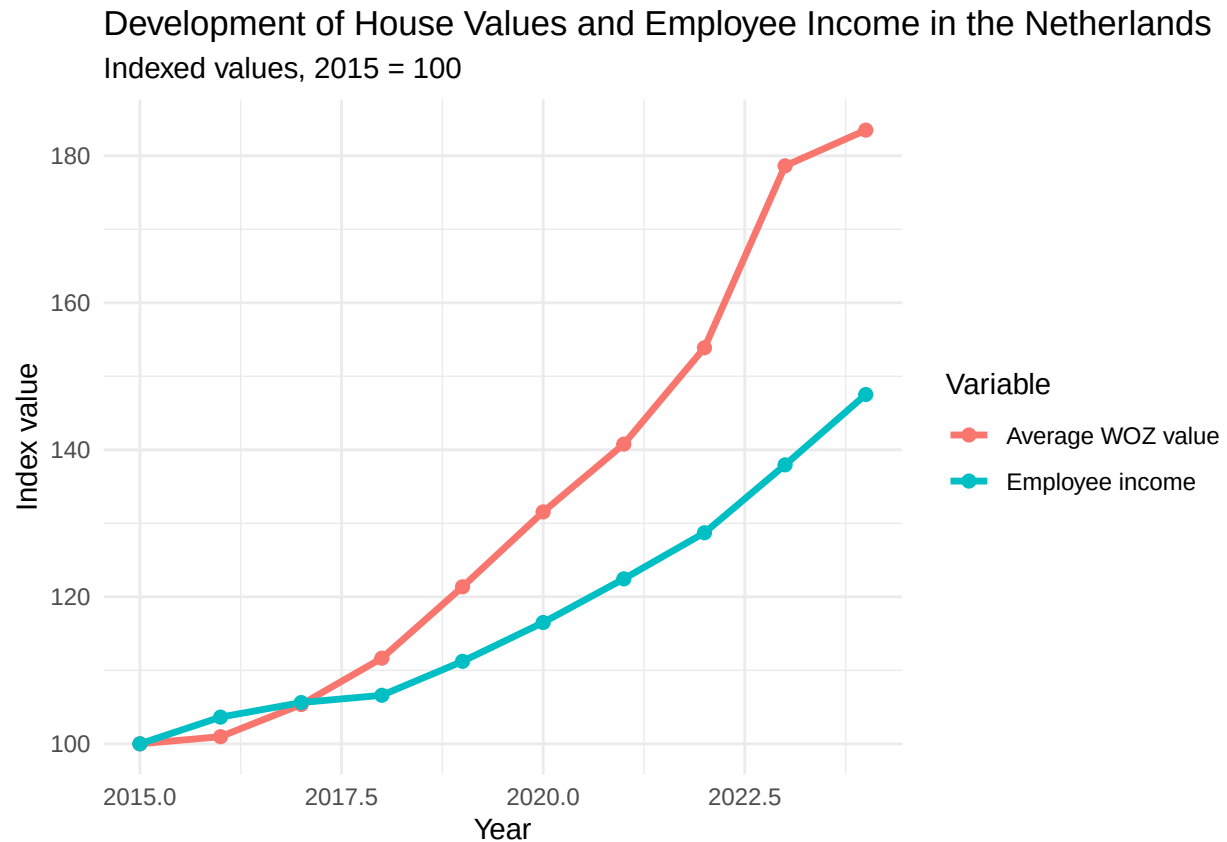
```
landelijk_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    WOZ_index = GemiddeldeWOZ / GemiddeldeWOZ[Year == 2015] * 100,
    Inkomen_index = InkomenWerknemer / InkomenWerknemer[Year == 2015] * 100
  ) %>%
  select(Year, WOZ_index, Inkomen_index) %>%
  pivot_longer(
    cols = c(WOZ_index, Inkomen_index),
    names_to = "Variable",
    values_to = "Index"
  ) %>%
  mutate(
    Variable = recode(
      Variable,
      WOZ_index = "Average WOZ value",
      Inkomen_index = "Employee income"
    )
  )

ggplot(landelijk_index, aes(x = Year, y = Index, color = Variable)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
```

```

title = "Development of House Values and Employee Income in the Netherlands",
subtitle = "Indexed values, 2015 = 100",
x = "Year",
y = "Index value",
color = "Variable"
) +
theme_minimal()

```



```

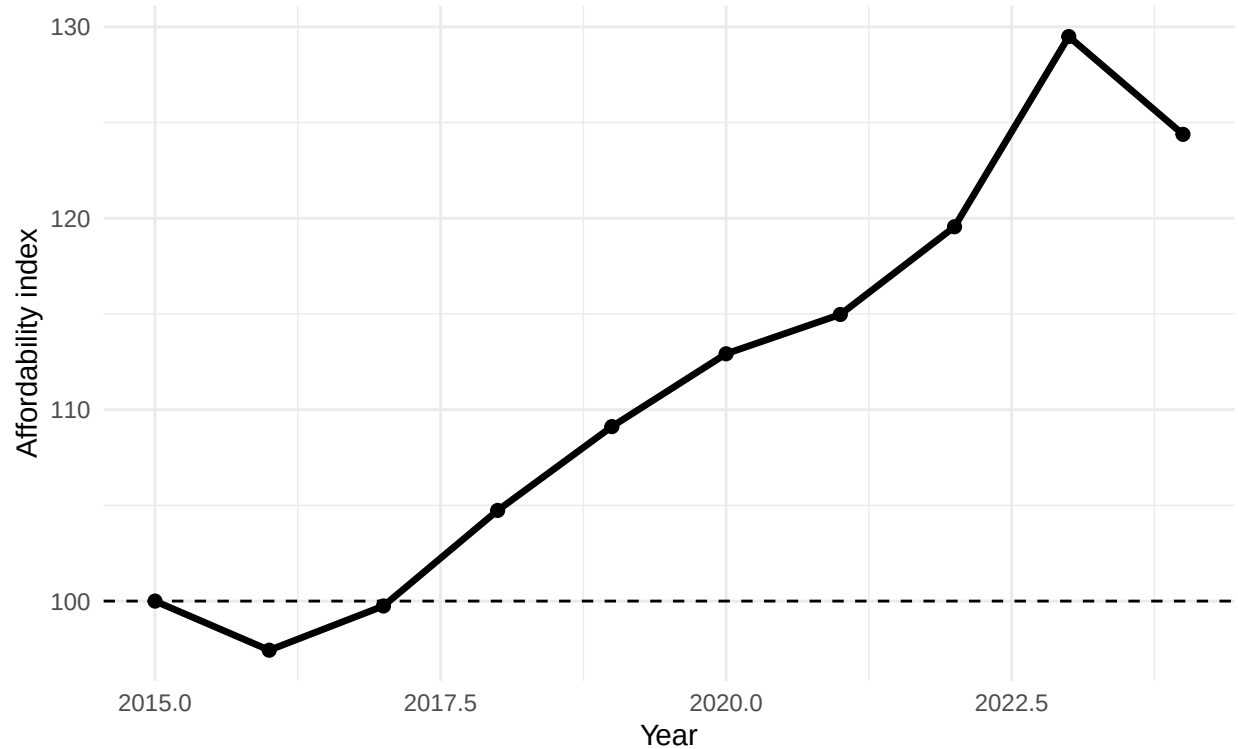
landelijk_affordability_index <- landelijk %>%
  arrange(Year) %>%
  mutate(
    AffordabilityIndex = AffordabilityRatio / AffordabilityRatio[Year == 2015] * 100
  )

ggplot(landelijk_affordability_index, aes(x = Year, y = AffordabilityIndex)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  geom_hline(yintercept = 100, linetype = "dashed") +
  labs(
    title = "Housing Affordability Index in the Netherlands",
    subtitle = "Affordability ratio indexed to 2015 = 100",
    x = "Year",
    y = "Affordability index"
  ) +
  theme_minimal()

```


Housing Affordability Index in the Netherlands

Affordability ratio indexed to 2015 = 100



```
urban_affordability <- raw_dataset %>%
  mutate(
    Year = as.integer(substr(Perioden, 1, 4))
  ) %>%
  filter(
    Year == 2024,
    startsWith(RegioS, "GM")
  ) %>%
  select(
    Year,
    RegioS,
    RegioS_label,
    GemiddeldeWOZWaardeVanWoningen_98,
    BronInkomenAlsWerknemer_141,
    ZeerSterkStedelijk_52,
    SterkStedelijk_53,
    MatigStedelijk_54,
    WeinigStedelijk_55,
    NietStedelijk_56
  ) %>%
  rename(
    GemiddeldeWOZ = GemiddeldeWOZWaardeVanWoningen_98,
    InkomenWerknemer = BronInkomenAlsWerknemer_141
  ) %>%
  mutate(
    AffordabilityRatio = GemiddeldeWOZ / InkomenWerknemer
```

```

) %>%
pivot_longer(
  cols = c(
    ZeerSterkStedelijk_52,
    SterkStedelijk_53,
    MatigStedelijk_54,
    WeinigStedelijk_55,
    NietStedelijk_56
  ),
  names_to = "UrbanizationType",
  values_to = "UrbanizationValue"
) %>%
group_by(RegioS, RegioS_label, Year) %>%
slice_max(UrbanizationValue, n = 1, with_ties = FALSE) %>%
ungroup() %>%
mutate(
  UrbanizationType = recode(
    UrbanizationType,
    ZeerSterkStedelijk_52 = "Very strongly urban",
    SterkStedelijk_53 = "Strongly urban",
    MatigStedelijk_54 = "Moderately urban",
    WeinigStedelijk_55 = "Weakly urban",
    NietStedelijk_56 = "Non-urban"
  ),
  UrbanizationType = factor(
    UrbanizationType,
    levels = c(
      "Very strongly urban",
      "Strongly urban",
      "Moderately urban",
      "Weakly urban",
      "Non-urban"
    )
  )
)

ggplot(urban_affordability, aes(x = UrbanizationType, y = AffordabilityRatio)) +
  geom_boxplot() +
  labs(
    title = "Housing Affordability by Urbanization Level",
    subtitle = "Municipalities grouped by dominant urbanization category, 2024",
    x = "Urbanization level",
    y = "WOZ-to-income ratio"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 25, hjust = 1)
  )

```

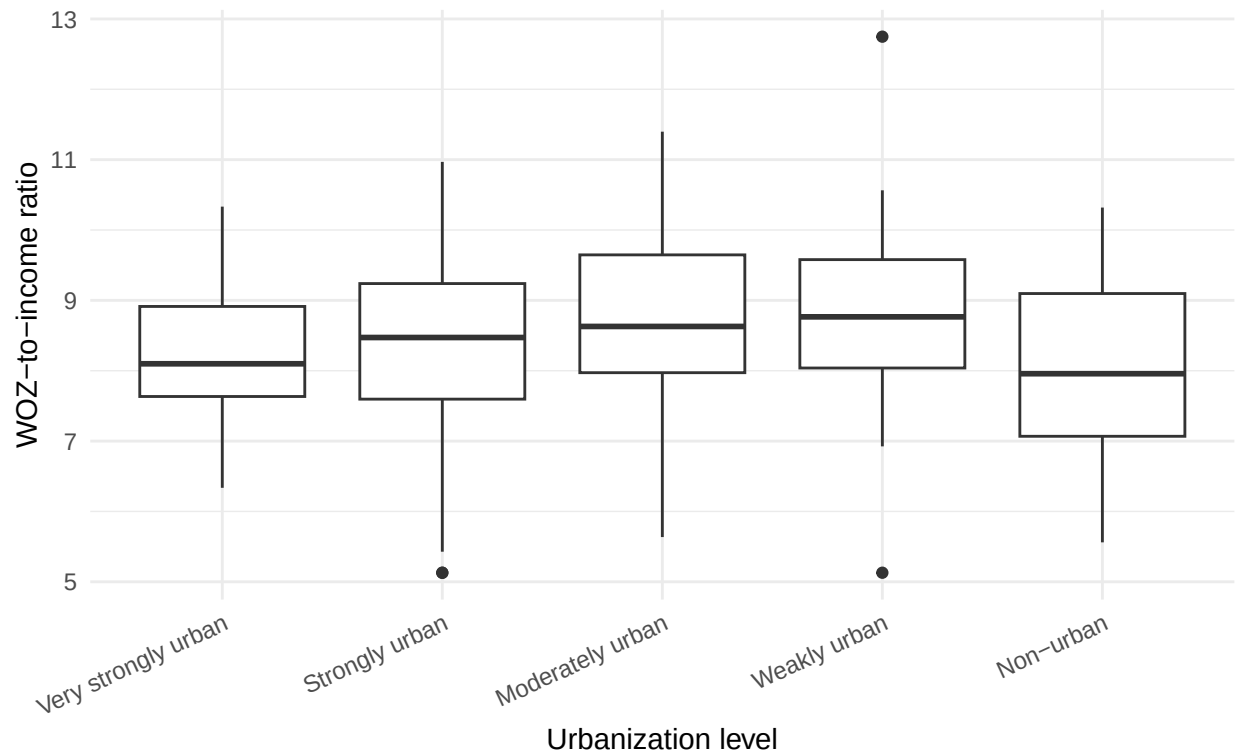
```

## Warning: Removed 408 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```

Housing Affordability by Urbanization Level

Municipalities grouped by dominant urbanization category, 2024



In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
#mean(dataset$percollege)
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

Variable 2

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.